

1. (a)

(i): L_2 norm: $\because x_i \sim (0, 1)$

$$\therefore E(x_i) = 0, \sigma^2 = E[x_i^2] = 1$$

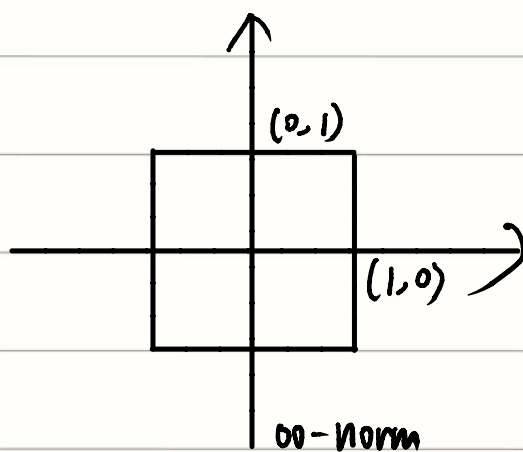
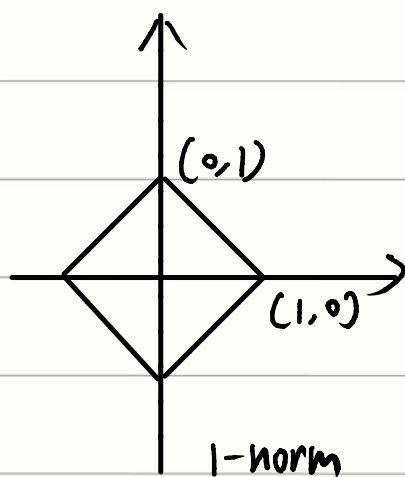
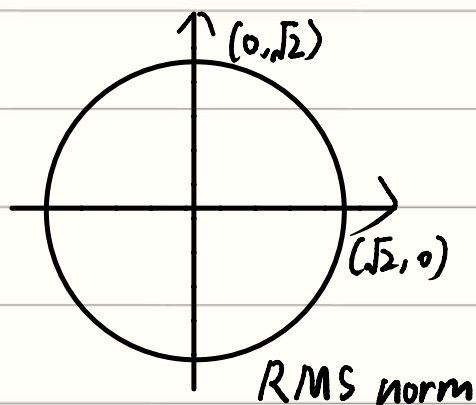
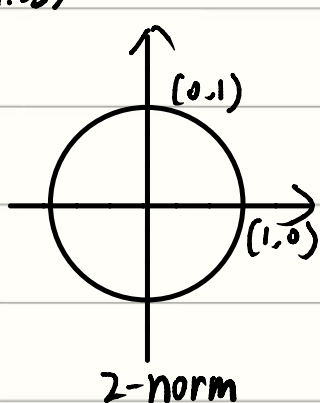
$$E[\|x\|_2] = E\left[\sqrt{\sum_{i=1}^d x_i^2}\right] \leq \sum_{i=1}^d \sqrt{E[x_i^2]} = \sqrt{d}$$

$$\therefore E[\|x\|_2] = O(\sqrt{d})$$

$$\because \|x\|_{RMS} = \frac{1}{\sqrt{d}} \cdot \|x\|_2$$

$$\therefore E[\|x\|_{RMS}] = O(1)$$

1. (b)



Less

is More

2.(a)

$$\text{Let } h(\Delta\theta) = g^T \Delta\theta + \frac{1}{2\alpha} \|\Delta\theta\|_2^2$$

$$h'(\Delta\theta) = g + \frac{2}{\alpha} \Delta\theta$$

$$\text{when } h'(\Delta\theta) = 0: g + \frac{2}{\alpha} \Delta\theta = 0$$

We can get $\Delta\theta = -\frac{\alpha}{2}g$, it's like gradient descent

$$2.(b): \because \|\Delta\theta\|_\infty = \max_i |\Delta\theta_i|$$

Let $M = \max_i |\Delta\theta_i|$, when can transfer the problem to:

$$u = \arg\min_{\Delta\theta} g^T \Delta\theta + \frac{1}{2\alpha} M^2 \text{ subject to } -M \leq \Delta\theta_i \leq M$$

In order to make $g^T \Delta\theta$ as small as possible, $\Delta\theta$ must be $-M \cdot \text{sign}(g)$

Substitute $\Delta\theta = -M \cdot \text{sign}(g)$ into $g^T \Delta\theta + \frac{1}{2\alpha} M^2$

$$\text{We can get } -M \cdot (g^T \cdot \text{sign}(g)) + \frac{1}{2\alpha} M^2$$

$$\because g^T \text{sign}(g) = \sum_{i=1}^n |g_i| = \|g\|_1$$

$$\therefore \text{We can transfer the problem to } \arg\min_M -M \|g\|_1 + \frac{1}{2\alpha} M^2$$

Similarly, take its derivative and then find the point where the derivative is zero

$$\text{We can get } M = \frac{\alpha}{2} \|g\|_1$$

Therefore, $\Delta\theta = -\frac{\alpha}{2} \|g\|_1 \text{sign}(g)$, it's like $\text{sign}(g)$ but not exactly the $\text{sign}(g)$

because $\text{sign}(g)$'s $\Delta\theta$ is $-\text{sign}(g)$, and this $\Delta\theta$ is $-\frac{\alpha}{2} \|g\|_1 \text{sign}(g)$

$$WW^T: d_{out} \times d_{in} \times d_{in} \times d_{out} \Rightarrow C \cdot d_{out}^2 d_{in}$$

Less

$$\{ I_{d_{out}} - WW^T: d_{out} \times d_{out} \Rightarrow C \cdot d_{out}^2 d_{in} \\ W: d_{out} \times d_{in}$$

is More

$$4(a): \text{ As for } \frac{1}{2} I_{d_{out}} \cdot W: t_1 = C \cdot d_{out} \cdot d_{out} \cdot d_{in} = C \cdot d_{out}^2 d_{in}$$

$$\text{ As for } \frac{1}{2} WW^T W: t_2 = C \cdot d_{out} \cdot d_{in} \cdot d_{out} + C \cdot d_{out} \cdot d_{out} \cdot d_{in} = 2C d_{out}^2 d_{in}$$

$$\text{ Therefore: } t = t_1 + t_2 = \cancel{3C d_{out}^2 d_{in}} \quad t = 2C d_{out}^2 d_{in}$$

$$4(b): P(w) = \frac{1}{2} w (I_{d_{in}} - W^T W)$$

Less

is More

$$b(a). \|y\|_{rms} = \frac{1}{\sqrt{d_2}} \sqrt{\sum_{i=1}^{d_2} (y_i)^2}$$

$$E[\|y\|_{rms}^2] = E\left[\frac{1}{d_2} \cdot \sum_{i=1}^{d_2} y_i^2\right]$$

$$\text{As for } y_i: y_i = \sum_{j=1}^{d_1} w_{ij} x_j$$

$\therefore w_{ij}$ and x_j are independent

$$\therefore E[y_i] = \sum_{j=1}^{d_1} E[w_{ij}] E[x_j] = 0$$

$$\text{Var}[y_i] = \sum_{j=1}^{d_1} E[(y_i - E[y_i])^2]$$

$$= d_1 E[y_i^2]$$

$$= d_1 E[w_{ij}^2 x_j^2]$$

$$= d_1 E[w_{ij}^2] E[x_j^2]$$

$$= d_1 \text{Var}(w_{ij}) \text{Var}(x_j)$$

$$= d_1$$

$$\therefore E[\|y\|_{rms}^2] = E\left[\frac{1}{d_2} \cdot \sum_{i=1}^{d_2} y_i^2\right]$$

$$= \frac{1}{d_2} \cdot d_2 \cdot \text{Var}(y_i)$$

$$= d_1$$

$\therefore$ In order to let $E[\|y\|_{rms}^2] = 1$, we should let $\text{Var}(y_i) = 1$

$\therefore \text{Var}(y_i) = d_1 \cdot \text{Var}(w_{ij}) \cdot \text{Var}(x_j)$ and we can not change x

What's more, $\text{Var}\left(\frac{1}{\sqrt{d_1}} w_{ij}\right) = \frac{1}{d_1} \text{Var}(w_{ij}) = \frac{1}{d_2}$

Therefore, we can multiply w by $\frac{1}{\sqrt{d_1}}$

Less

is More

$$\begin{aligned} 6(b). \quad y' &= \text{sign}(g_i \cdot x_i^T) x_i \\ &= \text{sign}(g_i) \cdot \text{sign}(x_i^T) \cdot x_i \\ &= \text{sign}(g_i) \cdot \|x_i\|_1 \end{aligned}$$

$$\|y'\|_{\text{RMS}} = \frac{1}{\sqrt{d_i}} \cdot \sqrt{\sum_{i=1}^{d_i} y_i^2}$$

$$E[\|y'\|_{\text{RMS}}^2] = \frac{1}{d_i} E\left[\sum_{i=1}^{d_i} y_i^2\right] = E[y_i^2]$$

$$\because y_i = \text{sign}(g_{ij}) \cdot \|x_i\|_1$$

$$\therefore y_i^2 = \|x_i\|_1^2 = \left(\sum_{k=1}^{d_i} |x_{ki}|\right)^2$$

$$\therefore E[\|y'\|_{\text{RMS}}^2] = E\left[\left(\sum_{k=1}^{d_i} |x_{ki}|\right)^2\right]$$

$$= d_i \cdot E[|x_{ki}|^2] + 2 \cdot C_{d_i}^2 \cdot E[|x_{k1}|] E[|x_{k1}|]$$

$$= d_i \cdot 1 + 2 \cdot \frac{d_i(d_i-1)}{2} \cdot \sqrt{\frac{\pi}{2}} \cdot \sqrt{\frac{\pi}{2}}$$

$$= d_i + \frac{\pi}{2} d_i(d_i-1)$$

$$= \Theta(d_i^2)$$

$\therefore$ We should multiply the update by $\frac{1}{d_i}$